

Avi Singh

AI/ML & Data Enthusiast — Software Developer — Winner, Smart India Hackathon '26

Bareilly, Uttar Pradesh, India — Open to Work (Remote, Hybrid, On-site)

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PROFESSIONAL SUMMARY

I build intelligent, end-to-end software solutions at the intersection of computer vision, deep learning, and practical application development. Currently pursuing my Master of Computer Applications (MCA) at the Central University of South Bihar, I focus on transforming data into functional, deployable systems. My technical approach prioritizes clean infrastructure and measurable impact. Recently, I led the development of *vision_ugv_autonomy*—a vision-based autonomous navigation pipeline for GPS-denied outdoor UGVs—which secured 2nd Place at Hackbhoomi 2.0 (Smart India Hackathon 2026 College Edition). Whether analyzing complex datasets with Python and SQL, engineering predictive models using PyTorch, or building full-stack AI pipelines, I thrive on taking projects from theoretical frameworks to production-ready deployments.

EDUCATION

Central University of South Bihar	2026 – 2028
Master of Computer Applications (MCA)	

Invertis University	2023 – 2026
Bachelor of Computer Applications (BCA)	

EXPERIENCE & INTERNSHIPS

AI-ML Virtual Intern	April 2025 – June 2025
<i>Google for Developers & AICTE (via EduSkills)</i>	

- Achieved Grade A in a rigorous 10-week virtual internship program focused on artificial intelligence and machine learning foundations.
- Developed and implemented practical machine learning models, translating theoretical concepts into applied algorithmic solutions.

FEATURED PROJECTS

vision_ugv_autonomy (Smart India Hackathon 2026)

- **Tech Stack:** Python, Computer Vision, Deep Learning
- Engineered a vision-based autonomous navigation system for outdoor Unmanned Ground Vehicles (UGVs) operating in GPS-denied environments.
- Developed real-time path detection algorithms to distinguish traversable terrain from hazards (rocks, ditches) and implemented visual localization to estimate vehicle position and orientation without satellite data.

AI E-Waste Classification System

- **Tech Stack:** PyTorch, ResNet50, Streamlit, Hugging Face
- Built a deep learning solution to automate the identification and categorization of 8 specific electronic waste types to aid responsible recycling.
- Optimized model training using a 50-epoch two-stage approach (transfer learning and fine-tuning) with a ResNet50 architecture.
- Deployed the interactive GUI via Streamlit on Hugging Face Spaces for real-time user image classification and confidence scoring.

Universal AI Media Pipeline & AI Resume Analyzer

- **Tech Stack:** Python, Natural Language Processing, Machine Learning
- Designed an automated pipeline for rich media processing, integrating various AI models to streamline data extraction and content generation.
- Developed a Resume Analyzer utilizing NLP to parse, evaluate, and score applicant data against specific job descriptions.

SKILLS

- **Languages & Development:** Python, SQL, Dart, JavaScript, HTML, CSS
- **AI & Machine Learning:** PyTorch, TensorFlow, Scikit-learn, Exploratory Data Analysis
- **Data Science & Analytics:** Pandas, NumPy, SciPy, Matplotlib, Seaborn, Power BI, Advanced Excel, Statistical Data Analysis
- **Tools & Infrastructure:** Docker, Git/GitHub, Linux CLI (Ubuntu, CentOS), Android Studio, VS Code, Hugging Face, Windows, macOS

HONORS & AWARDS

Winner – Hackbhoomi 2.0 (Smart India Hackathon 2026)

Invertis University College Edition

- Awarded second place out of competing university teams for developing a robust autonomous navigation pipeline for UGVs. Lead contributor for Team Gandharvas.

LICENSES & CERTIFICATIONS

- LLM For Young Developers - Foundational Course – *NASSCOM / FutureSkills Prime (with IndiaAI & Meta)*
- Data Analysis Using Python – *IBM Skills Network / Credly*
- Introduction to Generative AI – *Simplilearn SkillUp (Powered by Google Cloud)*
- Python 101 for Data Science – *Cognitive Class (Powered by IBM Developer Skills Network)*
- SQL and Relational Databases 101 – *Cognitive Class (Powered by IBM Developer Skills Network)*
- Getting Started with Artificial Intelligence – *IBM SkillsBuild*
- Ethical Hacking 101: Beginners Guide to Ethical Hacking – *Simplilearn SkillUp*
- Python for Absolute Beginners – *EC-Council Continuing Education*
- Selenium Automation Testing for Beginners – *EC-Council Continuing Education*
- Getting Started with Microsoft 365 – *LinkedIn Learning*